

Mohamed Ebraheem

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EDUCATION

University of Toronto

Bachelor of Science in Computer Science **GPA: 3.75**

Toronto, ON

Expected Apr 2028

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Software Design, Computer Organization, Machine Learning, Probability I & II, Linear Regression Analysis, Linear Algebra II, Calculus III

EXPERIENCE

Software Engineering Associate

Mathematics & Computational Sciences Society (MCSS) + DeerHacks

Aug 2025 – Present

Mississauga, ON

- Built and deployed a production **developer-facing platform** serving over **500 UofT students**, improving reliability and response accuracy through iterative monitoring and user feedback.
- Designed a secure **backend intent/action architecture** that eliminated direct LLM-to-SQL execution, reducing database security risk and centralizing authorization, auditing, and query validation.
- Implemented **rate limiting and abuse protection** for event subscription APIs, accounting for proxy-forwarded client IPs to prevent circumvention while maintaining reliable user access.
- Built a shared session system using **Redis and JWT authentication**, enabling scalable conversational context with documented tradeoffs for future expansion.

Web Development Lead

University of Toronto Autonomous Scale Racing (UTASR)

Oct 2024 – Present

Mississauga, ON

- Led a **6-member engineering team** to build and deploy a real-time telemetry web platform, delivering **biweekly feature releases** through structured task breakdown, code reviews, and ownership.
- Designed and implemented backend services in **Node.js and TypeScript**, processing **thousands of telemetry messages per session across distributed services** while maintaining clear service boundaries and extensibility.
- Built and optimized **low-latency TCP and WebSocket pipelines**, reducing end-to-end data delivery latency by **30%** through reconnect handling and edge-case hardening.
- Authored internal technical documentation and onboarding guides to support maintainability, knowledge sharing, and rapid contributor onboarding.
- Deployed and monitored production services on **AWS (EC2 and NGINX)**, achieving **99% uptime during testing sessions** with TLS, JWT-based authentication, and centralized logging.

PROJECTS

DiscoverMyUni

Django, Python, Web Scraping, ML-assisted Parsing

- Built and deployed **content ingestion and syndication pipelines** for a student learning platform serving **500 users**, aggregating events from multiple external sources.
- Improved event extraction and classification accuracy to **85%** by integrating ML-assisted parsing with structured validation and rule-based filtering.
- Collaborated in a **6-person agile team** using pull requests, code reviews, and sprint-based delivery to ship stable features under evolving requirements.

Network Intrusion Detection System

Python, XGBoost, Pydantic, scikit-learn, Zeek, Argus, PyTorch

- Developed an end to end supervised learning pipeline for flow based intrusion detection using the UNSW NB15 dataset with 175341 training flows and 82332 test flows, producing reproducible preprocessing and model artifacts.
- Trained and evaluated XGBoost models for binary detection and attack family classification, achieving 0.987 PR AUC and 0.983 ROC AUC for benign versus attack detection and returning top ranked categories for triage.
- Implemented strict feature contracts with Pydantic schemas and designing an offline ingestion pipeline from Zeek logs, Argus outputs, and PCAP files to support realistic evaluation beyond static CSV inputs.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, C#, JavaScript, TypeScript, SQL, R, Julia, Kotlin, Bash, MATLAB, HTML/CSS

Frameworks & Libraries: Spring Boot, React, Node.js, Express, Django, Flask, FastAPI, Next.js, JUnit

Machine Learning / AI: PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face, OpenCV, XGBoost, LightGBM, NumPy, pandas, Matplotlib, Seaborn, JAX, CUDA, OpenAI API, LangChain, Transformer Models

Databases: PostgreSQL, MySQL, MongoDB, Redis, Firebase, Supabase

Cloud & DevOps: AWS (EC2, S3, Lambda), Google Cloud Platform, Microsoft Azure, Docker, Kubernetes, NGINX

Developer Tools: Git, GitHub, Jenkins, TravisCI, Jupyter, VS Code, IntelliJ IDEA, PyCharm, Linux, Vim, Neovim

Systems & Security: Linux, TCP/IP, WebSockets, JWT, RBAC, TLS, Secure Coding Practices